Pythagoras' Theorem ⏱ ~25 min

NAME _____ DUE _____ SCORE / EFFORT _____

No calculator unless a question is marked 🖩. Show your working.

1 · This week

1. Find the unknown side of each right-angled triangle. Give exact answers (surds where needed).

a) shorter sides 16 cm and 30 cm _____

b) hypotenuse 25 cm, one side 7 cm _____

c) shorter sides 4 cm and 6 cm _____

d) hypotenuse 9 cm, one side 3 cm _____

WORKING

2. 🖩 Round your answers to Q1 c) and d) to 1 decimal place. c) _____ d) _____

3. Right-angled or not? Use the converse. For any that are not right-angled, say whether the largest angle is acute or obtuse.

a) 11, 60, 61 _____ b) 8, 14, 16 _____ c) 1.6 m, 3 m, 3.4 m _____

WORKING

4. Farm gate. A rectangular farm gate is 3.6 m wide and 1.5 m tall. A straight brace runs diagonally from one corner to the opposite corner. How long is the brace? *(Hint: 1.5 and 3.6 are both multiples of 0.3.)*

WORKING

5. Cartesian plane. P is the point $(-2, 3)$ and Q is the point $(4, -5)$.

a) Find the distance PQ. _____ b) Find the gradient of PQ. _____

WORKING

6. Rich problem — kite. Suppose a kite flier holds the string 1.5 m above level ground. The string is 50 m long and pulled straight, and the kite is directly above a point on the ground 30 m (horizontally) from the flier.

a) How high above the ground is the kite?

b) The wind changes and the kite is now directly above a point 40 m away (same string length). How far has the kite dropped?

WORKING

2 · Spaced revision

7.

a) **(Week 2)** Evaluate $\frac{2^5 \times 2^3}{2^4}$. _____

b) **(Week 4)** Increase \$240 by 15%. _____

c) **(Week 7)** Solve $3(x - 4) = x + 10$. _____

d) **(Week 22)** The mean of 4, 7, 9 and x is 8. Find x. _____

e) **(Week 17)** A fair six-sided die is rolled. What is the probability of rolling a prime number? _____

WORKING

3 · Think deeper / At home

8. The rope-stretcher's trick. Builders and surveyors have long used a 3–4–5 triangle to check right angles. With someone at home and a tape measure, test a corner (a table, a doorway, a rug or a book): mark 30 cm along one edge and 40 cm along the other, then measure between the marks.

a) Record your three measurements and say whether the corner is right-angled, acute or obtuse.

b) Explain to your helper why this works, using the words *converse* and *hypotenuse*. Write one sentence about what they found surprising or asked.